Haemoglobin

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Introduction

- The main function of red blood cell
  - Transfer of O_2 from lungs to tissue
  - Transfer of CO_2 from tissue to lungs
- To accomplish this function red cells has haemoglobin (Hb)
- Each red cell has 640 million molecules of Hb

Introduction

- Haemoglobin (Hb), protein constituting 1/3 of the red blood cells
- Synthesis begins in proerythroblast
  - 65% at erythroblast stage
  - 35% at reticulocyte stage
- Two parts
  - Haem
  - Globin

Synthesis of Haemoglobin (Hb)

- Haem & globin produced at two different sites in the cells
  - Haem in mitochondria
  - Globin in polyribosomes

Structure of hemoglobin

- Hemoglobin is a globular protein made up of four subunits, each of which contains a polypeptide chain called globin and a heme group.

  The globin tertiary structure comprises a helical structures joined by non-helical segments. Four such globins are arranged together, giving rise to the spherical quaternary structure of hemoglobin

  Hemoglobins are classified into different types, depending on the combination of the two sets of globin units. Most of the hemoglobin present in adult humans comprises 2 α globins and 2 β globins

Structure of hemoglobin

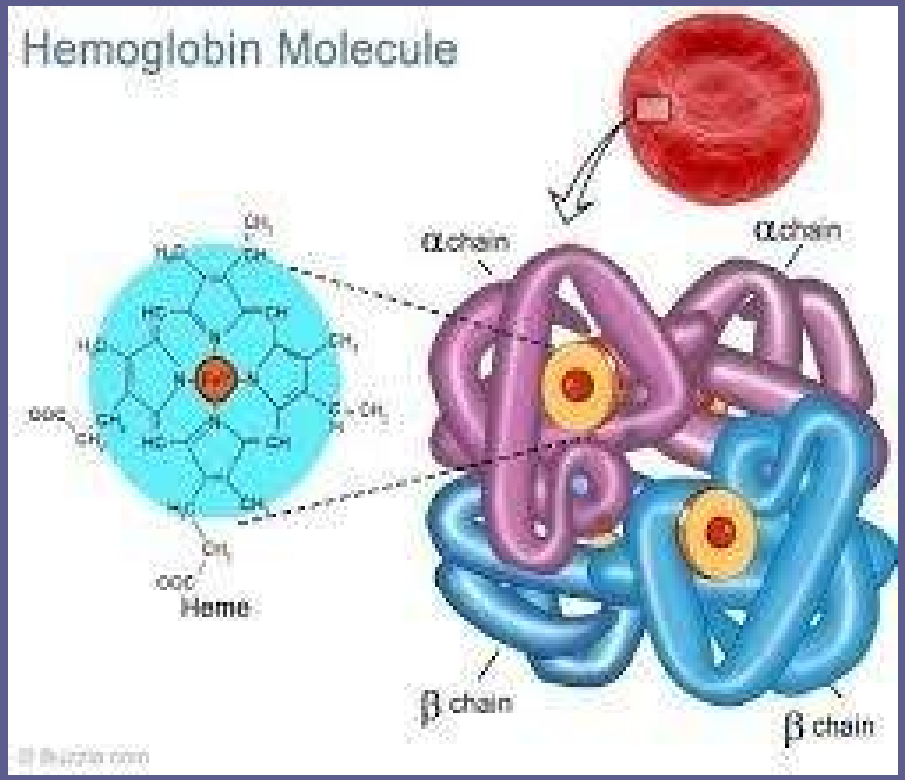
- The heme group bound to each globin is an organic macromolecule with an iron at the center. It serves as the prosthetic group of hemoglobin. The prosthetic group of a protein refers to any tightly-bound non-protein entity, that is essential for the structural and functional integrity of the protein.

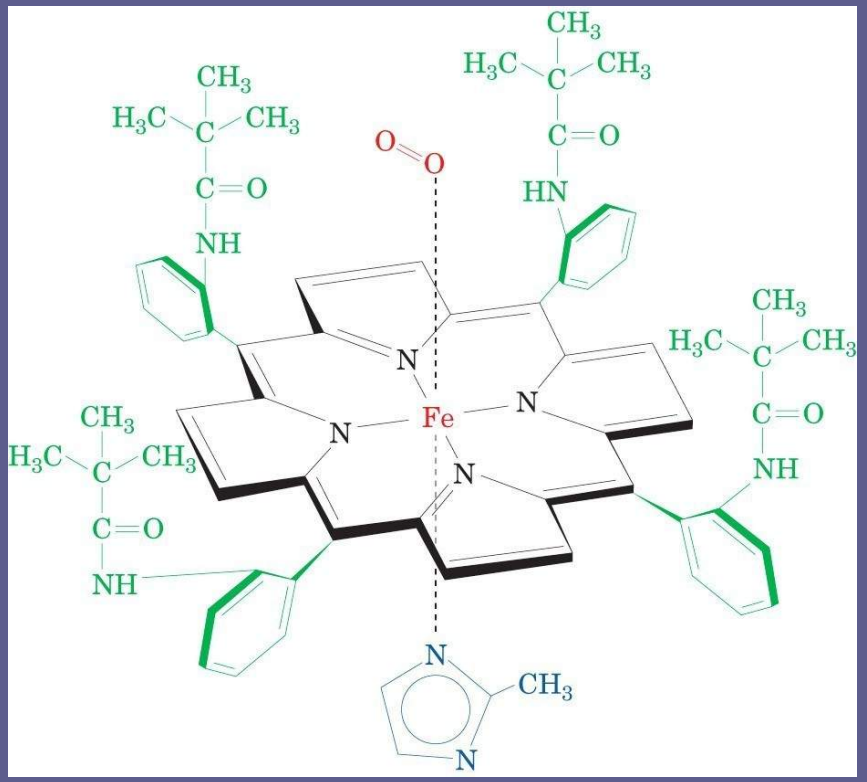
  The heme group comprises a structure called the porphyrin ring, which is formed by the combination of four heterocyclic rings called pyrroles. An iron ion (Fe^{2+}) is present at the center of this structure, and is bound to the nitrogen atoms of the four pyrrole rings. It is this central iron which provides the reversible binding to oxygen and carbon dioxide molecules.

Structure of hemoglobin

made up of four subunits each of which contains globin and a heme group

The heme group: Fe^{2+}–porphyrin complex with bound O_2

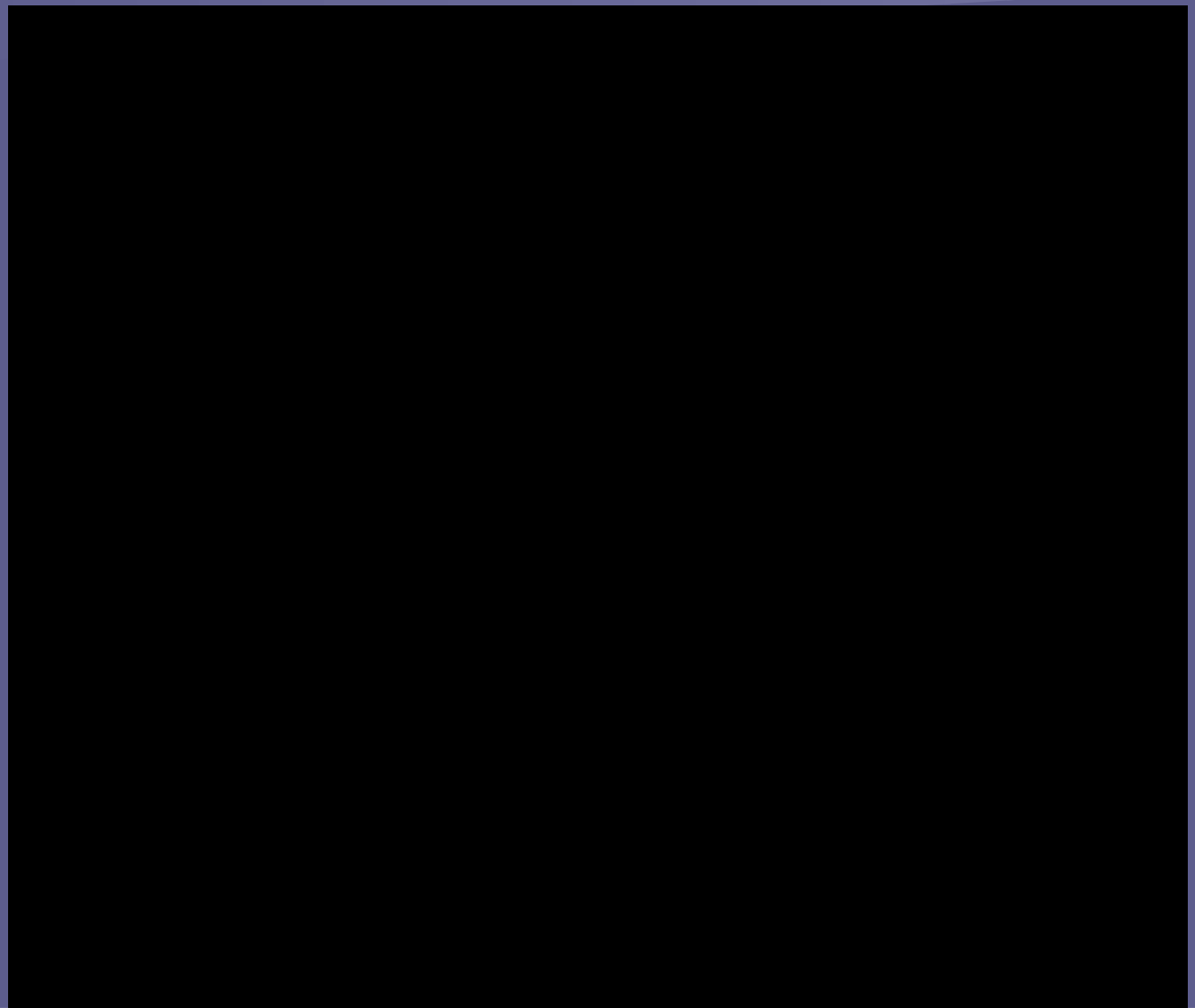




HB structure animation



Oxygen transport



Functions

- Oxygen transport is the main function of hemoglobin, and more than 98% of the oxygen in blood is carried through hemoglobin. In addition, it also transports carbon dioxide released by peripheral tissues to the lung tissues. During this process, the hemoglobin macromolecule undergoes conformational changes due to the binding and unbinding of oxygen and carbon dioxide.and it include :

1_ Oxygen Pickup

- 2_ Oxygen Delivery

3_ Carbon Dioxide Pickup

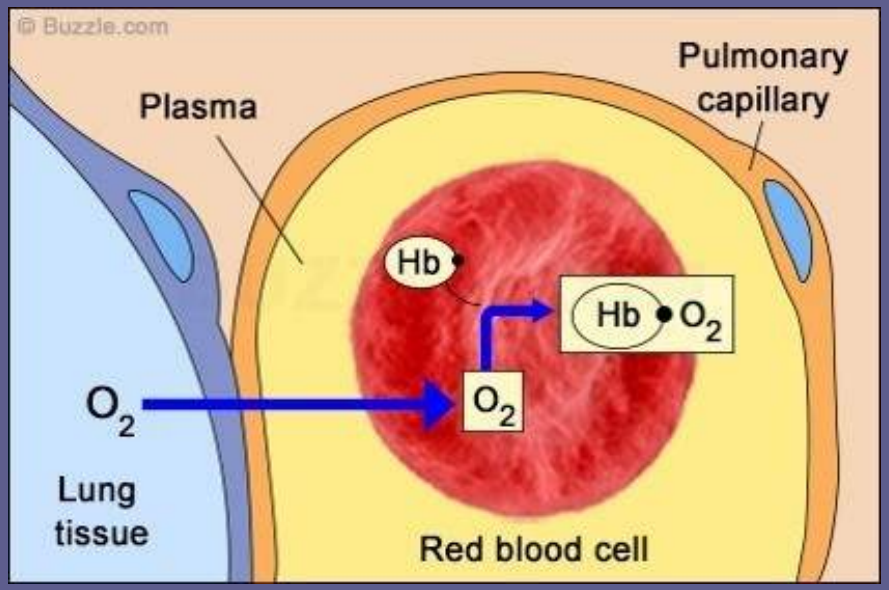
4_ Carbon Dioxide Delivery

Functions

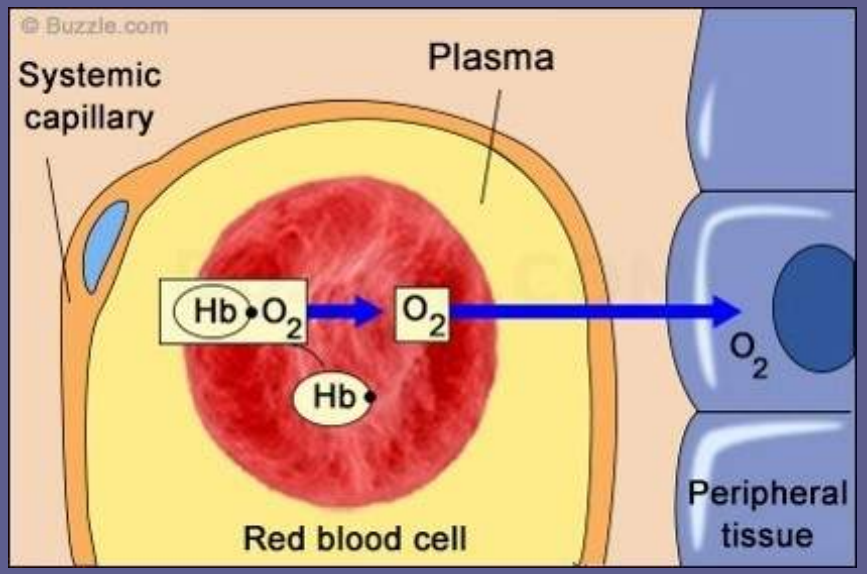
- Notes
  1_Oxygenation not oxidation
  2_One Hb can bind to four O2 molecules
  3_Less than .01 sec required for Oxygenation..

Functions

Oxygen Pickup

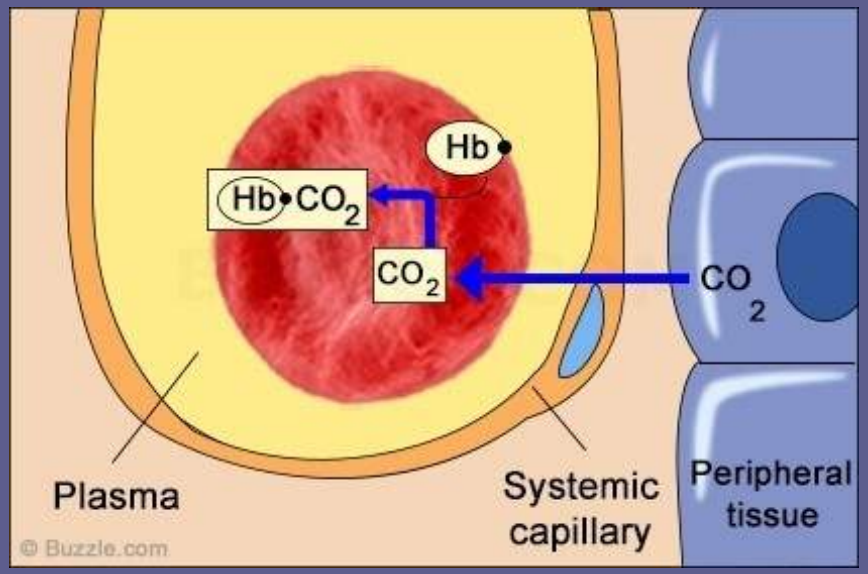


Oxygen Delivery

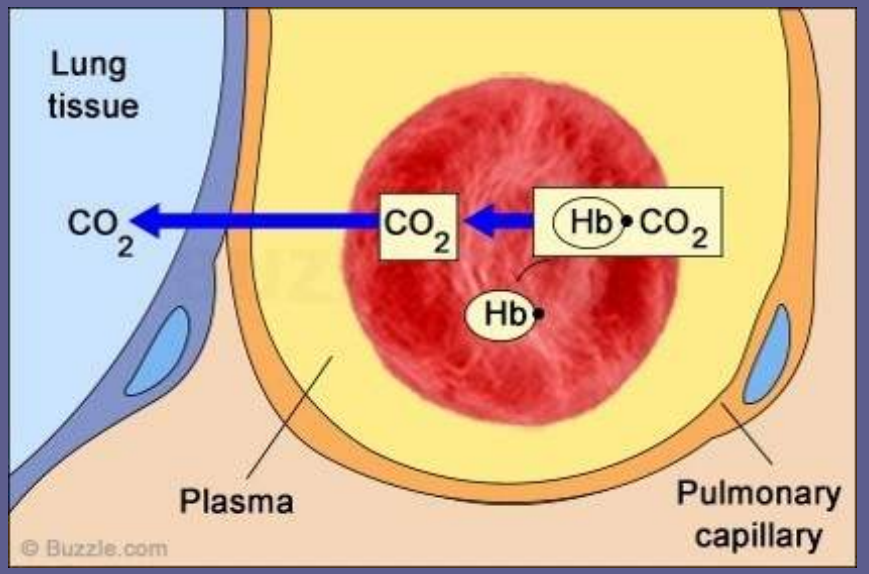


Functions

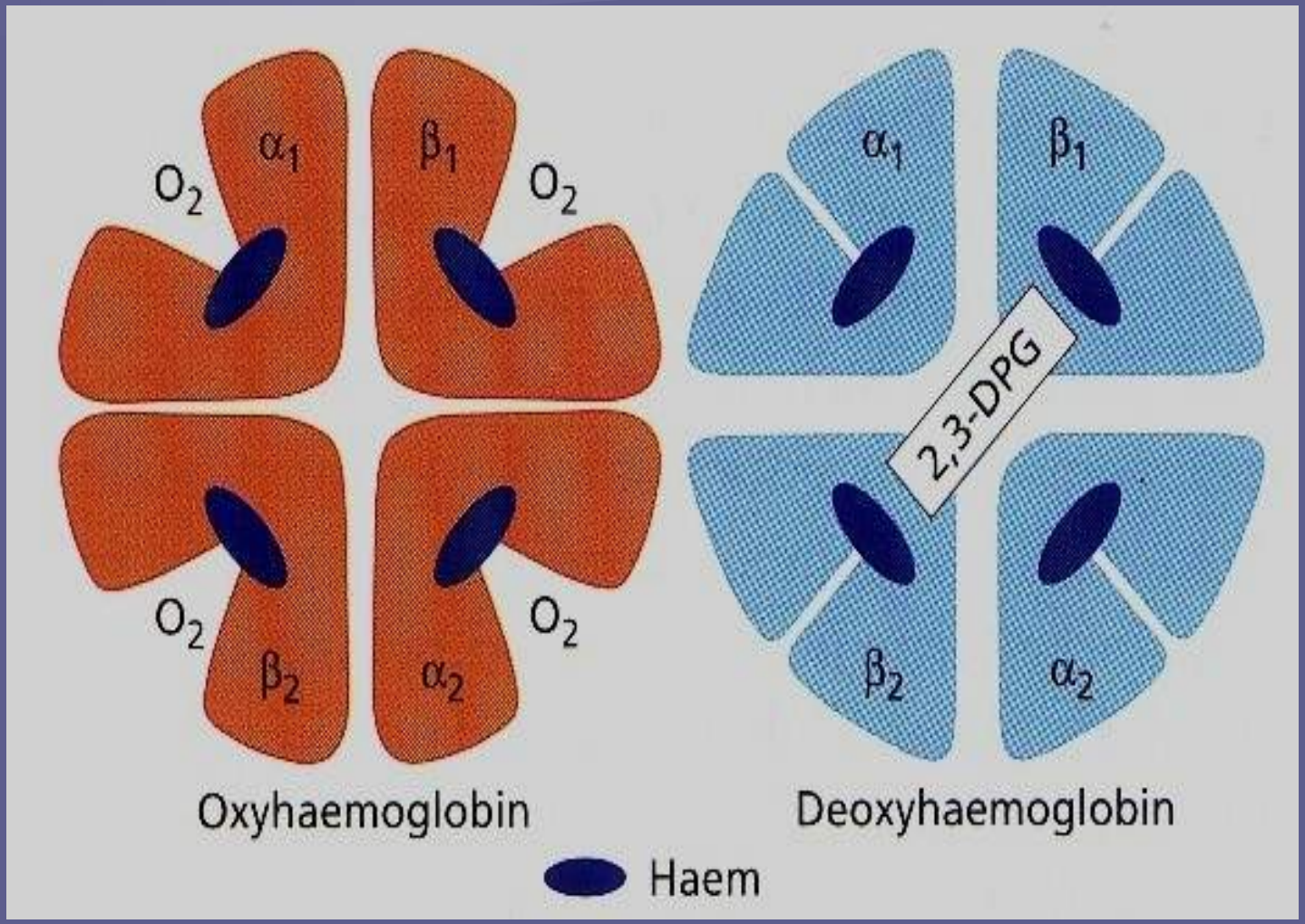
Carbon Dioxide Pickup



Carbon Dioxide Delivery



Oxy & deoxyhaemoglobin



Factors affecting oxygen binding

- In order to function most efficiently, hemoglobin needs to bind to oxygen tightly in the oxygen-rich atmosphere of the lungs and be able to release oxygen rapidly in the relatively oxygen-poor environment of the tissues
- Essentially, hemoglobin is an allosteric protein that has more than one shape and can undergo conformational changes in its structure based on environment conditions

Factors affecting oxygen binding

- The ability of hemoglobin to take up oxygen molecules in the lungs and then release them in the tissues is regulated by several factors both within the hemoglobin molecule itself and through external chemical factors
- allosteric effectors:
  - pO_2 (partial oxygen pressure)
  - pH of the environment
  - pCO_2 (partial carbon dioxide pressure)
  - Availability of 2,3-bisphosphoglycerate

Factors affecting oxygen binding

1. **Heme-heme interactions**

- One of the biggest regulators of the oxygen affinity of the hemoglobin is the presence of oxygen itself
- In the lungs where the oxygen levels are high, the hemoglobin has a higher affinity for oxygen and this affinity increases disproportionately with the number of molecules it already has bound to it

Factors affecting oxygen binding

- In other words, after the oxyhemoglobin binds one molecule of oxygen its affinity for oxygen increases until the hemoglobin is fully saturated.
- In the same way, the deoxyhemoglobin has a lower affinity for oxygen and this affinity decreases disproportionately with the number of molecules it already has bound .

Factors affecting oxygen binding

- Thus, the loss of one oxygen molecule from the deoxyhemoglobin lowers the affinity for the remaining oxygen. This regulation is known as Cooperativity

- Cooperativity is essential to the functioning of the hemoglobin because it allows the oxyhemoglobin to carry the maximum amount of oxygen to the tissues and then allows the deoxyhemoglobin to release the maximum amount of oxygen into the tissues .

Factors affecting oxygen binding

- Net effect is that the affinity of Hb for the last oxygen bound is 300 times greater then its affinity for the first oxygen bound.
- This effect is called heme- heme interaction.

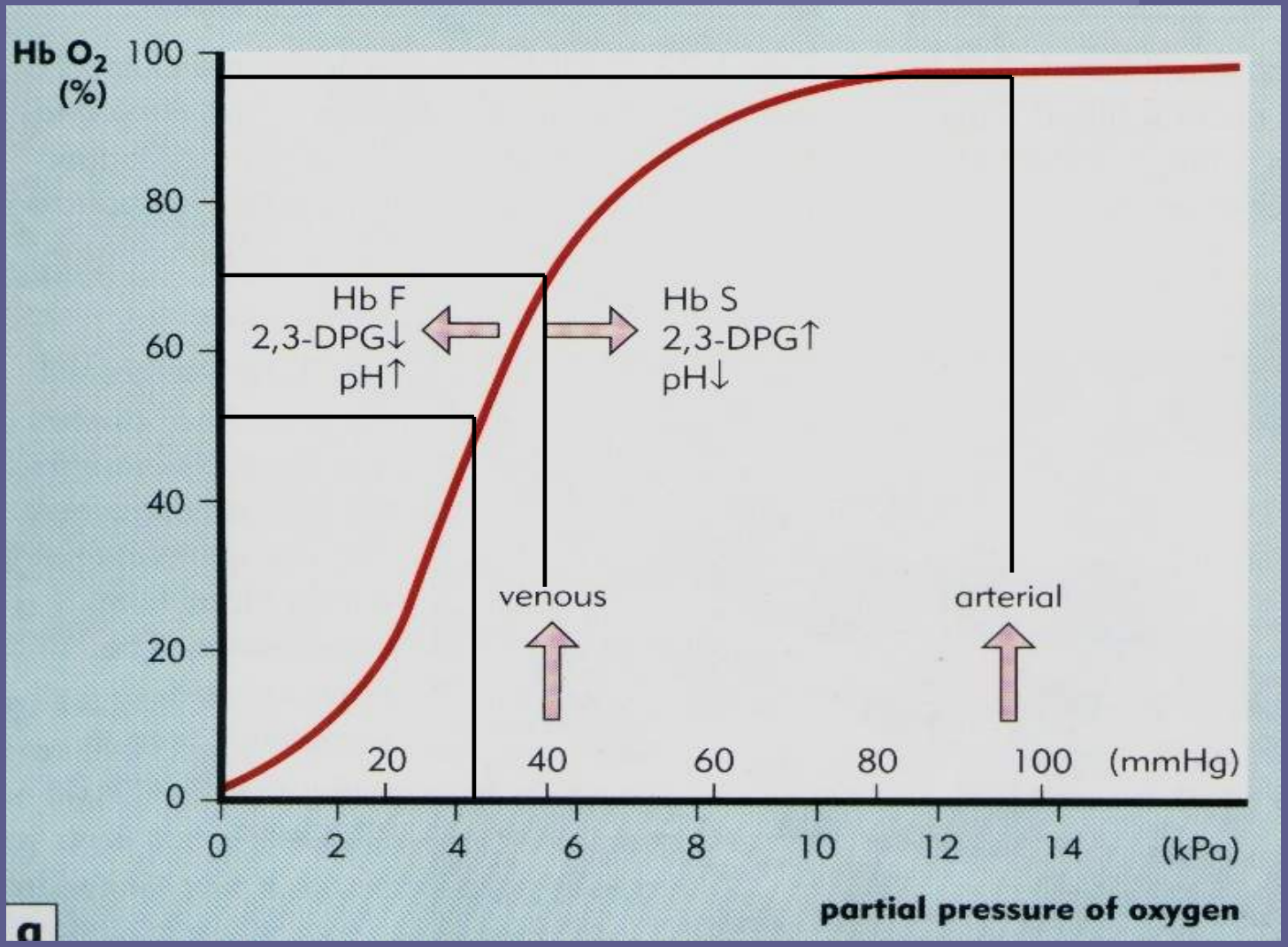
Oxygen-haemoglobin dissociation curve

- O_2 carrying capacity of Hb at different Po_2
- Sigmoid shape
  - Binding of one molecule facilitate the second molecule binding
  - P_{50} (partial pressure of O_2 at which Hb is half saturated with O_2) 26.6mmHg

Hb-oxygen dissociation curve

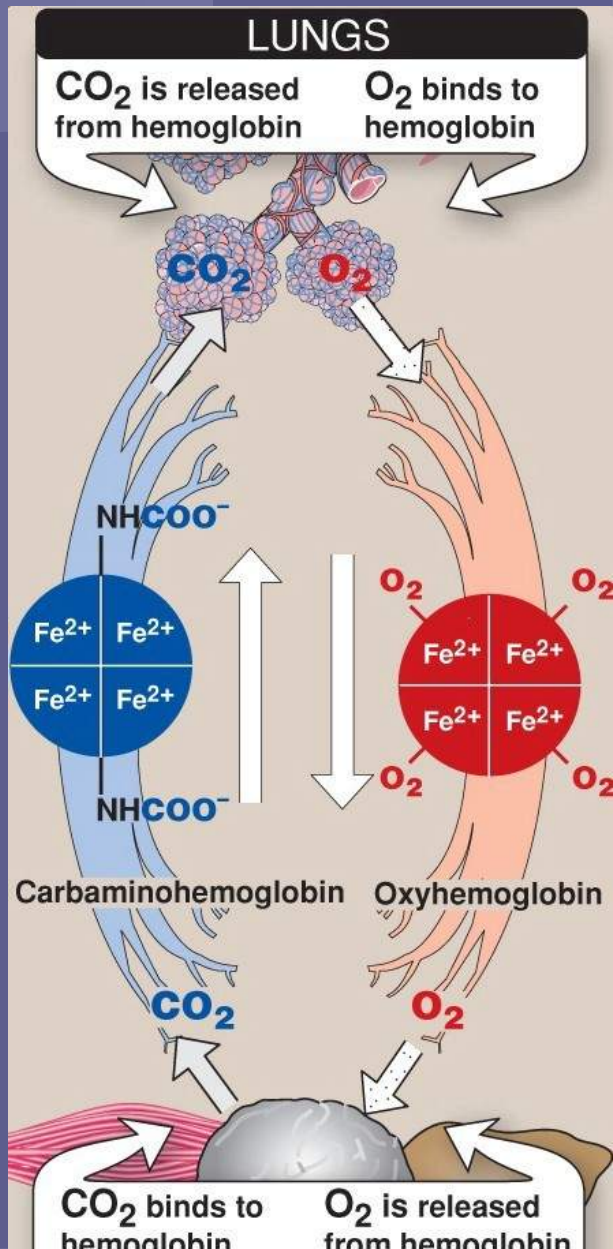
- The normal position of curve depends on
  - Concentration of 2,3-DPG
  - H^+ ion concentration (pH)
  - CO_2 in red blood cells
  - Structure of Hb

Hb-oxygen dissociation curve

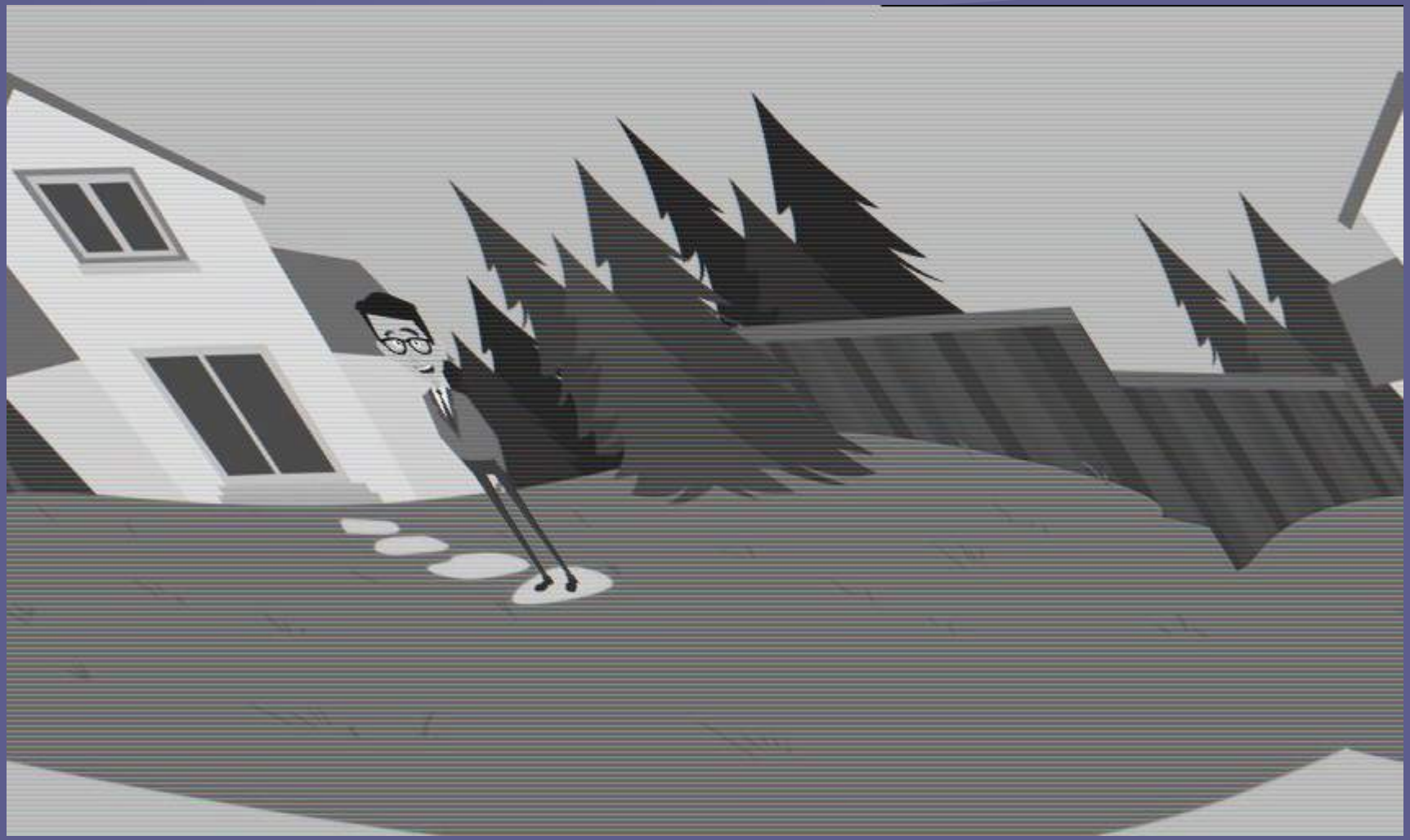


P_{50}

- Indicates affinity of Hb to O_2
- P_{50}(mm Hg): the pressure at which Hb is 50% saturated with O_2
- High affinity ◊ slow unloading of O_2
- Low affinity ◊ fast unloading of O_2
- Lung pO_2 is 100 mm ◊ Hb saturation 100%
- Tissue pO_2 is 40 mm ◊ Hb saturation reduces



Dissociation curve



Factors affecting oxygen binding

2. **Bohr Effect**:

- When CO_2 is released into the blood from the tissues it acidifies the blood by increasing the concentration of hydrogen ions.
- This lowering in pH causes the oxygen affinity of hemoglobin to decrease, which is known as the Bohr effect

Factors affecting oxygen binding

- The molecular basis behind the Bohr effect is that the T structure of hemoglobin binds hydrogen more readily than the R structure, so under a condition of low pH (high hydrogen ion concentration) the T structure, which has a decreased oxygen affinity, dominates.
- Bohr effect is because of the ionizable groups plus histidine side chain

Factors affecting oxygen binding

- When the conc of ions increase these groups are charged and are able to form ionic bonds. These bonds stabilize the T form and so the affinity for oxygen decreases
- Bohr effect causes the shift to the right in the oxygen dissociation curve

Sources of Protons:

a. Hydrogen ions and CO_2 more in the metabolically active tissues.

b. Organic acids produced during anaerobic metabolism (role of carbonic anhydrase.

Factors affecting oxygen binding

Mechanism of Bohr effect:

- Bohr effect reflects that the deoxyform of Hb has got a greater affinity for protons than oxyHb.
- It is because of the ionizable groups
- When conc of protons increases these groups become charged and form ionic bonds.
- These bonds stabilize the deoxyform of Hb and produces a decrease affinity for oxygen.

Factors affecting oxygen binding

3. 2,3 – Bisphosphoglycerate :

- 2,3 - bisphosphoglycerate is an allosteric effector that changes the oxygen affinity of hemoglobin by binding to the hemoglobin itself.
- It decreases the oxygen affinity of hemoglobin by binding to deoxy form. This stabilizes the taut structure.

$$HbO_2 + 2,3BPG \rightleftharpoons Hb\text{-}2,3BPG + O_2$$

Factors affecting oxygen binding

- 2,3 BPG binds to a pocket formed by two beta- globin chains in the center of the tetramer.
- Pocket has got positively charged amino acids that forms ionic bonds with 2,3BPG.
- When O_2 binds then 2,3 BPG is expelled
- 2,3 BPG causes the shift to the right in the oxygen dissociation curve.

Factors affecting oxygen binding

- 2,3 BPG levels are increased in chronic hypoxia (high altitudes, COPD) and chronic anemia.
- 2,3 BPG absent in fetal Hb.
- Role in transfused blood (acid citrate dextrose) (Inosine hypoxanthine ribose).
- 2,3 BPG is essential for the normal transport function of Hb. Storing of blood in acid citrate medium decreases 2,3 BPG and the affinity of Hb for oxygen increases

Factors affecting oxygen binding

- Fetal Hb has got a gamma globin chain, this has got less positive charge so does not bind 2,3 BPG.
- This allows the HbF to facilitate the transfer of oxygen from the maternal blood to the fetal blood.

Factors affecting oxygen binding

4. **Binding of CO_2 :**

- CO_2 has a similar effect on the hemoglobin, but instead of binding to the heme molecule like oxygen, CO_2 binds to the N-terminus of the alpha globin molecule.
- The CO_2 binds better to the globin in the T structure, so the release of oxygen in the tissues by the T structure of hemoglobin facilitates the uptake of CO_2

Factors affecting oxygen binding

- CO_2 is mostly transported in the form of bicarbonate.
- Some is transported by Hb. It is called carbamino hemoglobin or carbamate, because it is attached to the uncharged α-amino groups

Ligand binding

- Besides the oxygen ligand which binds to hemoglobin in a cooperative manner, hemoglobin ligands also include competitive inhibitors such as carbon monoxide (CO) and allosteric ligands such as carbon dioxide (CO_2).

Competitive

- Hemoglobin's oxygen-binding capacity is decreased in the presence of carbon monoxide because both gases compete for the same binding sites on hemoglobin, carbon monoxide binding preferentially in place of oxygen.
- The binding of oxygen is affected by molecules such as carbon monoxide (CO) (for example from tobacco smoking, car exhaust and incomplete combustion in furnaces). CO competes with oxygen at the heme binding site.

- Hemoglobin binding affinity for CO is 200 times greater than its affinity for oxygen, meaning that small amounts of CO dramatically reduce hemoglobin's ability to transport oxygen
- When hemoglobin combines with CO, it forms a very bright red compound called carboxyhemoglobin, which may cause the skin of CO poisoning victims to appear pink in death, instead of white or blue
- When inspired air contains CO levels as low as 0.02%, headache and nausea occur; if the CO concentration is increased to 0.1%, unconsciousness will follow. In heavy smokers, up to 20% of the oxygen-active sites can be blocked by CO

affinity for cyanide(CN^-), sulfur monoxide (SO), nitrogen dioxide(NO_2), and sulfide(S^{2-}), including hydrogen sulfide (H_2S).

All of these bind to iron in heme without changing its oxidation state, but they nevertheless inhibit oxygen-binding, causing grave toxicity

Allosteric

- Carbon dioxide occupies a different binding site on the hemoglobin. Carbon dioxide is more readily dissolved in deoxygenated blood, facilitating its removal from the body after the oxygen has been released to tissues undergoing metabolism. This increased affinity for carbon dioxide by the venous blood is known as the Haldane effect

- Through the enzyme carbonic anhydrase, carbon dioxide reacts with water to give carbonic acid, which decomposes into bicarbonate and protons:
- $CO2 + H2O \rightarrow H2CO3 \rightarrow HCO3^- + H^+$
- Hence blood with high carbon dioxide levels is also lower in pH (more acidic). Hemoglobin can bind protons and carbon dioxide which causes a conformational change in the protein and facilitates the release of oxygen

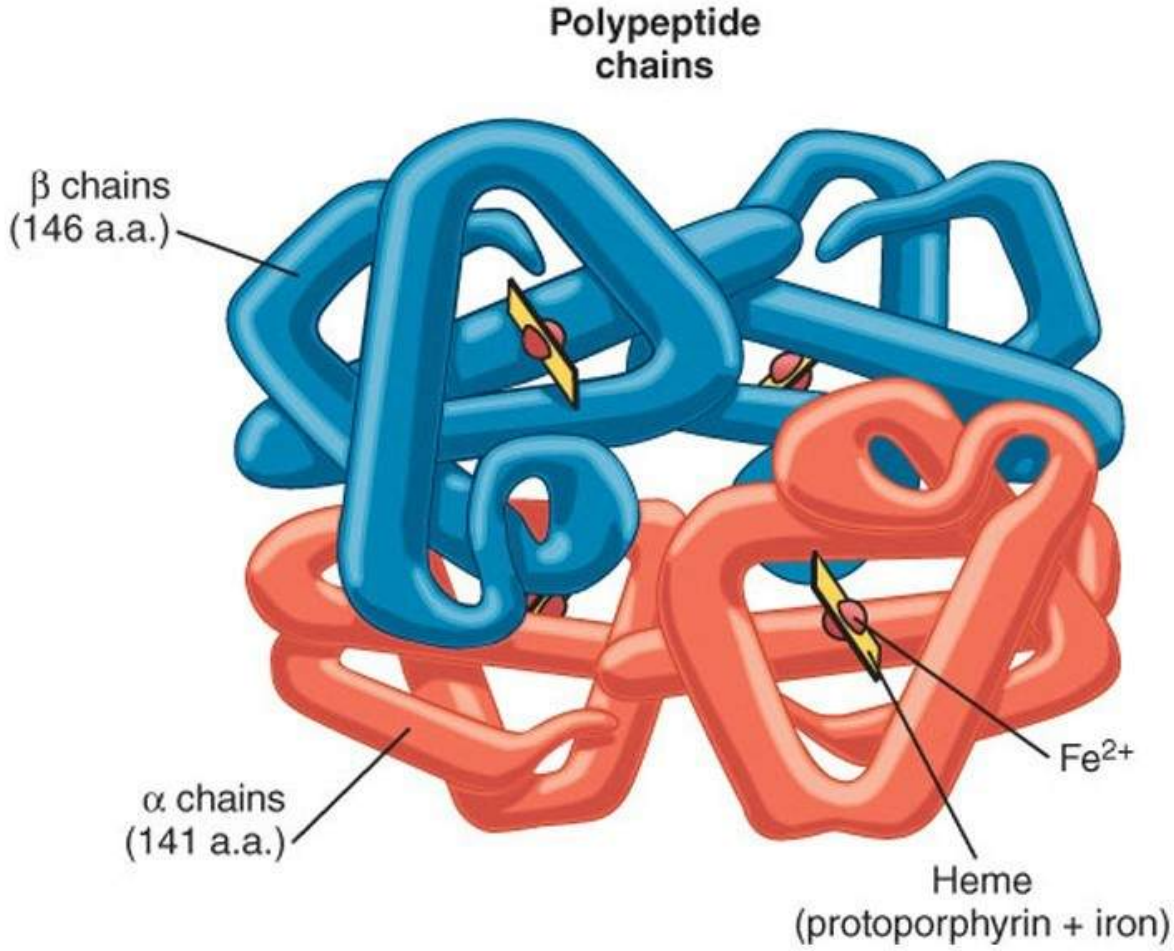
$$HbO_2 + H^+ _____ HbH + O_2$$

Types of Hb

Normal	HbA (97%)
	HbA2 (2%)
	HbF (1%)
	HbA_{1c}
abnormal	Carboxy Hb
	Met Hb
	Sulf Hb

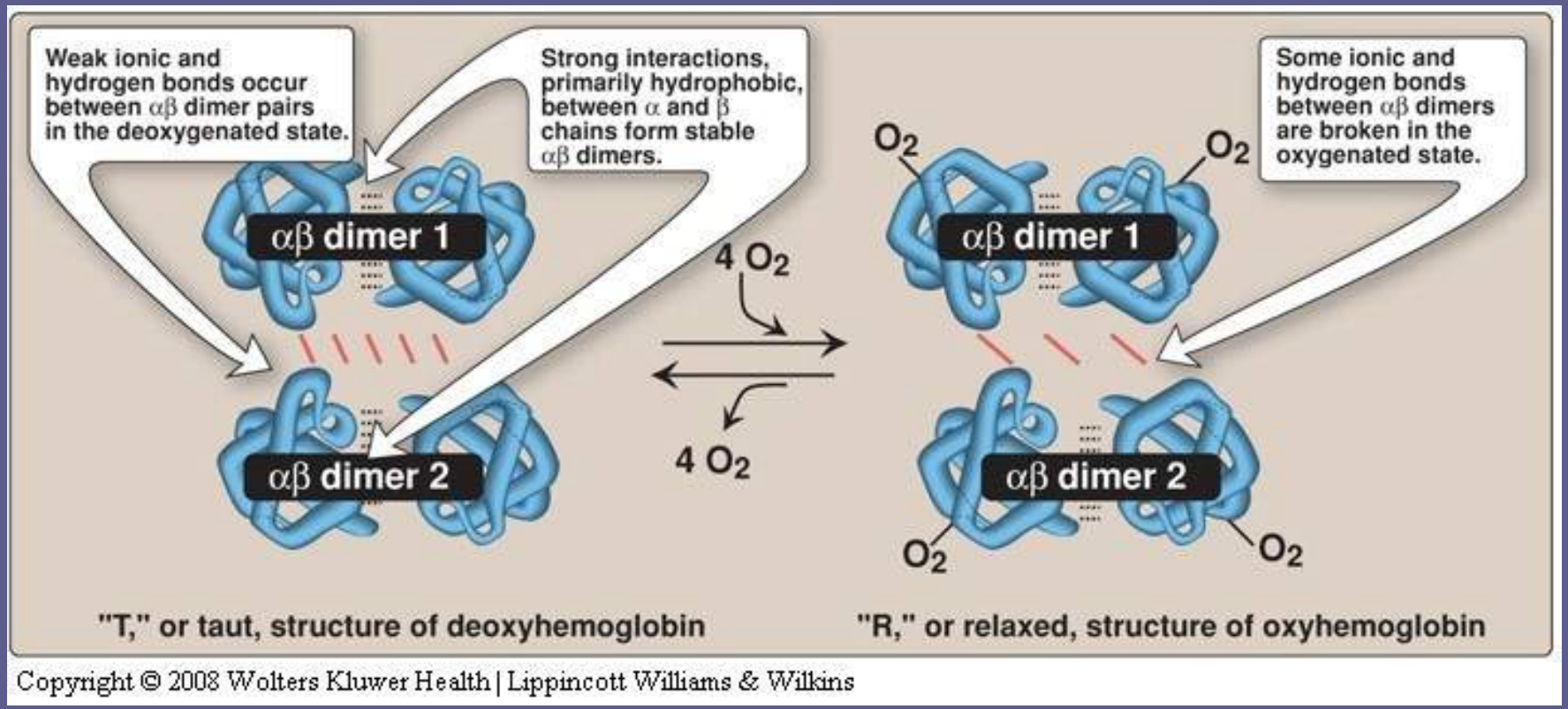
Hemoglobin A (HbA)

- Major Hb in adults
- Composed of four polypetide chains:
  - Two α and two β chains
- Contains two dimers of $\alpha\beta$ subunits
- Held together by non-covalent interactions
- Each chain is a subunit with a heme group in the center that carries oxygen
- A Hb molecule contains 4 heme groups and carries 4 moelcules of O_2



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HbA structure



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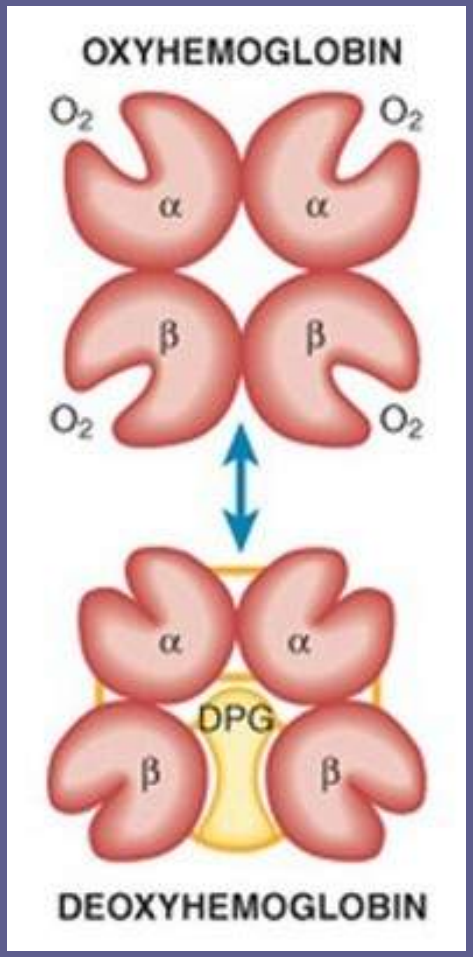
T-form of Hb

_The deoxy form of Hb

_Tout form

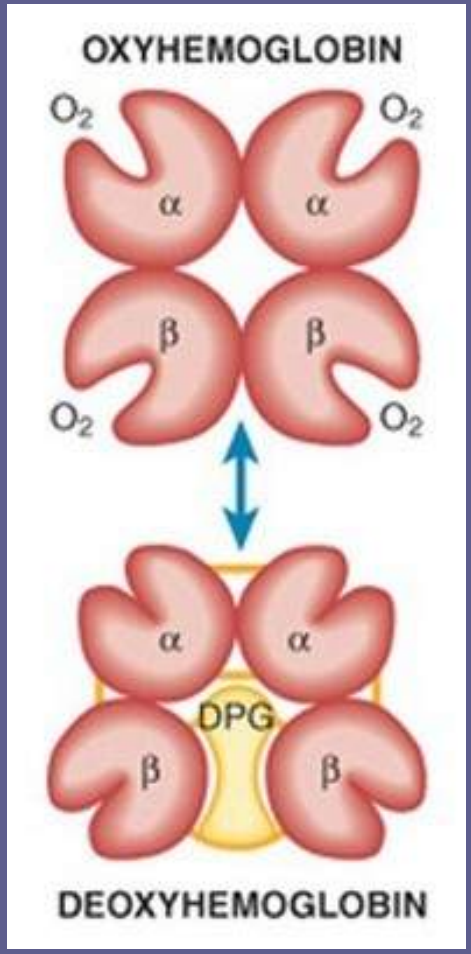
_The movement of dimers is constrained

_Low-oxygen-affinity form



R-form of Hb

_The oxygenated form of Hb
Relaxed form
_The dimers have more freedom of movement
_High-oxygen-affinity form



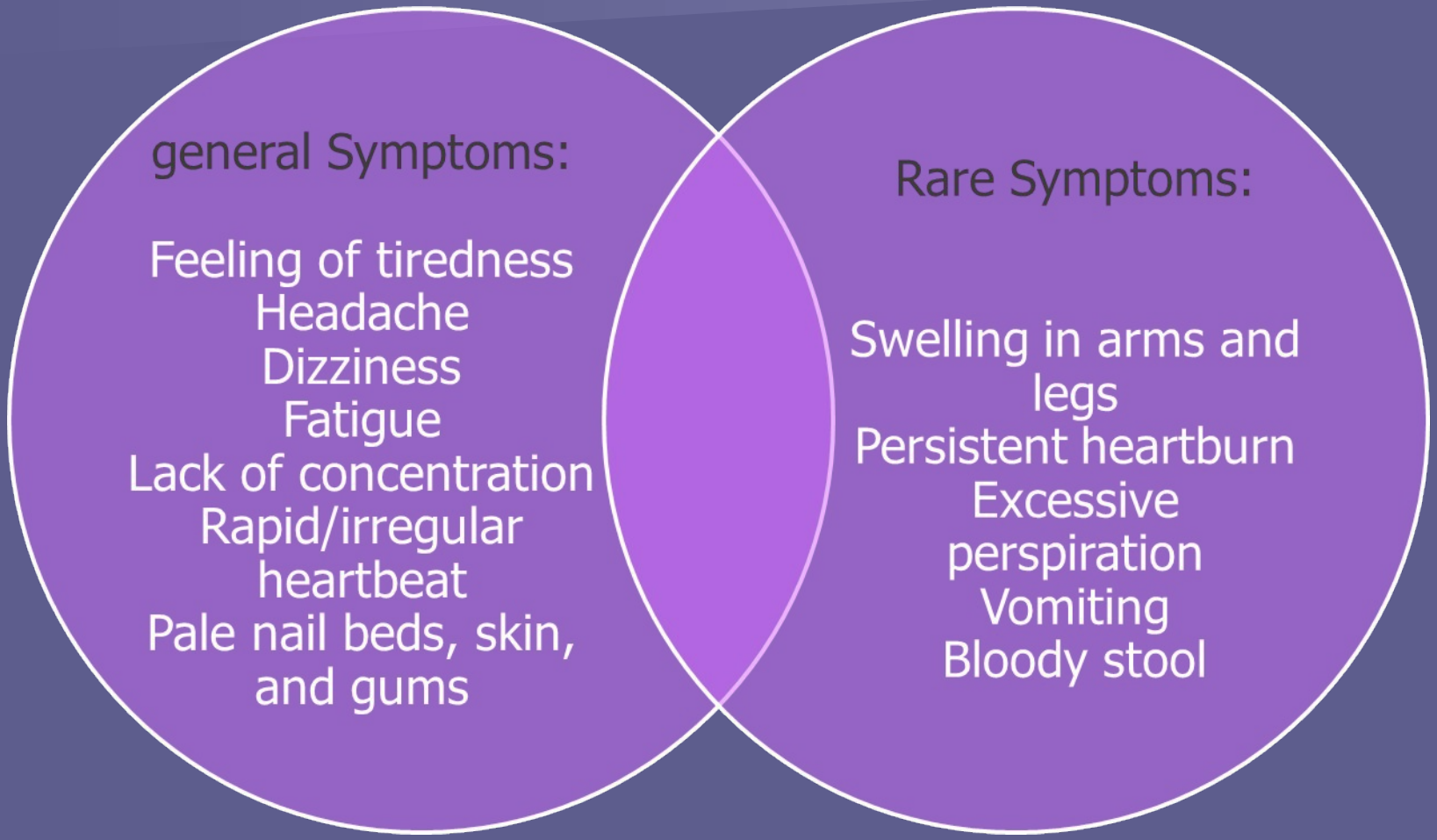
Normal Hemoglobin Levels

- It (Hb) is measured with a unit called gm/dl.
- Newborn Babies 17- 22
- Children 11-13
- Adults (Male) 14-18
- Adults (Female) 12-16
- Elderly (Male) 12.4-14.9
- Elderly (Female) 11.7-13.8

Causes of Low Hemoglobin Levels

- Nutritional Deficiency
- Inability to Absorb Iron
- Blood Loss
- Lowered Production of Red Blood Cells
- Destruction of Blood Cells

Symptoms of Low Hemoglobin Levels



How to Increase Hemoglobin Levels

- Food Sources and Supplements
- Avoid iron blockers
- Reduce intake of foods with oxalic acid
- Avoid foods with gluten
- Increase intake of vitamin C

